Conformity review v3

Structural Aliasing in Patch-Based Time-Series Foundation Models — re-audit after the δ_O fix, the rewritten §7, Appendix F and the disclosed row filter

What changed since v2

- ▲ **δ_O is settled on the contrast d.** The estimand is now stated on the orientation Model A is actually fitted to, where a mitigation is a *positive* slope. Fixed in `four.tex`, `tab:bayesModels`, `tab:bayesDecisions` (rule and caption), `bayesian_appendix.tex` (both places), `eight.tex` (rule, verdict, agent paragraph, future work), `ten.tex` (two places), `tab:riskRegister` (T4 and the caption), and in `bayesian_analysis.ipynb` (the probability, the verdicts table, and a new markdown note). The verdict moved from **refuted** to **inconclusive**: the slope leans the way $M1$ predicts on both sets ($\Pr(\delta_O > 0) = 0.964$ and 0.977), but the LOO leg is unevaluated and Model A fails its convergence checks. The M1 row in the notebook now requires *both* legs before it will print SUPPORTED.
- ▲ **The row filter is disclosed** in §7, beside the two exclusions that were already there, with its definition, the fact that it is evaluated on the whole triplet rather than the candidate arm, and its direction — excluding those rows makes an effect easier to find, not harder, so an inconclusive behavioural result is not an artefact of it.
- ▲ **The three Appendix E pages are in `figures/`** and the placeholders no longer fire.
- ▲ **§7 is rewritten** with a clean four-part structure and the design rationale for retraining every geometry; ▲ **Appendix F** now documents the construction, streaming data and optimisation. Between them these close the two largest documentation gaps of v2.
- ▲ The stray `[]` in Appendix A are gone. ▲ The `six.tex:3` broken sentence is gone. ▲ The claim that the sweep shows TSMixup and KernelSynth views has been corrected to what the appendix actually contains. ▲ The unsupported frequentist-concordance sentence is gone with the old §8.
- **§8 Bayesian Results is empty** while the results are re-run. Everything below reads the document as it stands; the verdicts that depend on numbers are marked **PENDING** rather than failed.

Verdict at a glance

#	Requirement	Verdict	Change from v2
1	Tokenisation operator, patch identity	**PARTIAL**	Unchanged. The operator is in `one.tex`, the identity and observability derivations in `two.tex`, which is where they belong; the standing gap is *indistinguishable*, not the split across sections.
2	f_s, P, S, f; lock class	**MET**	Unchanged.
3	Semantic layer and its exposure	**PARTIAL**	Unchanged.
4	Falsifiable hypotheses	**PARTIAL**	Improved: $M1$'s rule is now oriented and both its legs are stated wherever it is applied. θ_{lock} still has no pre-registered row.
5	Controlled experiments, isolation	**MET**	Strengthened by the new §7 and Appendix F.
6	Behavioural and representational	**PENDING**	Both levels are designed and instrumented; the results section is empty.
7	Bayesian analysis	**PENDING**	The specification is now internally consistent and the filter is disclosed; the numbers are pending.
8	Mitigation strategy	**MET**	Verdict corrected; the geometric argument was always the strong half and is untouched.
9	Agentic AI	**MET**	Unchanged.
10	NIST AI RMF risk assessment	**PARTIAL**	T4 corrected. The RMF functions are still only a table column.

Word count. 7,080 text + 132 headings = **7,212 words** with §8 empty. The old §8 was 701 words, so expect roughly 7,900 once it is rewritten, against the 8,800 limit. That leaves about 900 words of headroom — enough for what is missing below, but no longer generous. Executive summary 389 words, within its 400 cap.

Standing blockers

B1. The executive summary is still Lorem ipsum (`sections/summary.tex`), on page 1.

B2. "CHATGPT ACADEMIC PROPOSAL" is still in the body, `sections/eight.tex:21`, with lines 22–24 a paraphrased duplicate of 17–19. It sits inside the section carrying requirements 8 and 9, and it renders. Delete 21–24.

B3. §8 is empty. Every number in §9, §10 and `tab:riskRegister` currently cites a section that says nothing — $e^{\theta_{lock}} = 1.27$, $\kappa_S = 1.000$, $\Pr = 0.997$, and the δ_O posterior quoted in §9 itself. The register caption still claims every entry is "anchored to a posterior reported in **??**". This resolves itself when the results land, but nothing can be submitted until it does.

B4. When §8 is rewritten, three things have to be right this time: a verdict for $H3$-location (v2 found none stated); θ_P reported, since the rule needs it; and elpd_diff $\pm$ dse for every LOO comparison §7 promises — five of them, including $M1$'s second leg, which is what now decides $M1$. The notebook prints all of these.

B5. The convergence gate. §7 still says "Only a posterior that passes those checks is carried into the three-way rule". If B and C come back below their diagnostics again and you still want to report them, the sentence has to be softened to permit a qualified verdict — state the rule you will actually follow before the numbers go in, not after.

B6. No pre-registered rule for θ_{lock}. `tab:bayesDecisions` has no row for the representational estimand, yet a $\log 1.2$ threshold was applied to it. Add the row now, before the refit, or the claim that "every threshold is fixed before a posterior is inspected" (`four.tex`) is false again.

Findings by requirement

1 — Tokenisation operator PARTIAL

Taking the split across §2 and §3 as intended — `one.tex` defines the operator, `two.tex` derives what happens to it — the remaining gaps are substantive rather than editorial:

- *"Indistinguishable" is never formalised.* Only exact equality is derived. No ε-neighbourhood, no bound relating a detuning Δf to $\|t_{k+1} - t_k\|$ — while `ontology.tex:132-140` depends on tolerances $\varepsilon_S, \varepsilon_P$ that are asserted, and `six.tex` picks $\delta \leq 0.25 f_s/S$ by fiat. *Two sentences in §3 would close this: the first-order expansion of $t_{k+1} - t_k$ in Δf, and the observation that the control offset is chosen to sit outside that neighbourhood.*
- *Sufficiency only.* `eq:general_periodicity` assumes global invariance $x[n+S] = x[n]\ \forall n$; the exact condition for $t_{k+1} = t_k$ is local, over $m = 0, \dots, P-1$. For the non-sinusoidal signals §3 claims to cover, the difference is real.
- *Overclaim at `two.tex:260-262`*: "neighbours stay distinct unless $S = P$" is false on $\mathcal{F}_S \cap \mathcal{F}_P$ — e.g. $P = 16, S = 8, f = f_s/8$. The correct statement is made properly twenty lines earlier; only the summary sentence generalises it wrongly.

2 — Lock class and degeneracies MET

Derivation verified step by step. Two optional additions: the token-space degeneracy class is strictly larger than $\mathcal{F}_{\text{lock}}$ ($t_{k+j} = t_k$ at every *rational* cps, e.g. $P = S = 16$, $f = f_s/32$), and the existence condition $c < S/2$, $k < P/2$ is unstated. `tables/flockDomains.tex` and `tables/patchStride.tex` are still never `\input`, so $\mathcal{F}_{\text{lock}}$ is never instantiated numerically anywhere in the document.

3 — Semantic layer PARTIAL

Unchanged from v2. The concepts, states and relations are there; the clause the brief actually stresses — how aliasing affects *recognition* of those concepts — is three sentences, and no `Forecast` class exists in the ontology even though every claim about the agent runs through a forecast. *One class, one object property and a worked case of a missed `RampEvent` would move this to MET, at a cost of maybe 150 words.* Also standing: Z is used in `eq:observability-state` and never defined; `introduction.tex:21` calls the layer an OWL2 ontology when what is delivered is prose plus four tables.

4 — Falsifiable hypotheses PARTIAL

(i), (ii) and (iii) are covered with estimands and numeric thresholds, and $M1$'s rule is now correctly oriented and consistently two-legged. Standing:

- No pre-registered rule for θ_{lock} (B6).
- $H2$ is confounded with $H1$-behavioural and this is still unacknowledged: σ_ϕ is the spread of phase offsets on the same response where no deficit was found, so $\sigma_\phi \approx 0$ is close to guaranteed. *One caveat sentence in the rewritten §8 defuses it.*
- $H3b$: `four.tex` says the design supplies the patch series; the old §8 said it was not identified. If that recurs, reconcile the two.

5 — Controlled experiments MET

The strongest part of the report, and stronger than in v2. Isolation from classical sampling is argued four independent ways, `tab:hfModels` verifies arithmetically, and §7 now states the geometry rationale, the retraining rationale, the f_s/telemetry correspondence and the protocol order in four clean parts, with the full training recipe in Appendix F — including the two deliberate departures from the published recipe (100k steps, 10k shuffle buffer) applied uniformly so they stay outside the contrast. That last point is exactly the right thing to say. Remaining:

- *The phase design is still unspecified*: "the non-redundant alignments available at each frequency" leaves the number varying with frequency and untabulated. Of the two axes the brief names, patch parameters get a table and temporal offsets get a clause. *One sentence with the count would close it.*
- *The "full set of twenty-two" is still not listed* (it was in the old §8, and about half the reported numbers come from it). Note the released code disagrees: `probe_lib.py` defines 22 and then removes $S \in \{5, 15, 28\}$, leaving 17, which `collect.py` fits. $480 \bmod 5 = 480 \bmod 15 = 0$, so this is not the stride-boundary rule — either the count or the code is wrong.
- *Structural nit*: the sampling/band paragraph, the signal-set paragraph and the new filter paragraph all sit under the **Retraining** heading. A **Signals and paired observations** subsubsection before them would fix the outline in one line.

6 — Behavioural and representational PENDING

The design carries both levels and contrasts them rather than merging them, which is the report's best idea. Appendix E now renders its three pages, so the descriptive evidence base is visible for the first time. What is missing is the results section itself (B3) and, within it, θ_P, the $H3$-location verdict and the LOO numbers (B4). The spectral *comparison* of Appendix E is still three "Planned extension" subsections plus a `TO DO` comment at `sweep_appendix.tex:155`. *Either finish it with the decomposition figures you now generate, or cut the three subsections — unexecuted work presented in a final report reads worse than absent work.*

7 — Bayesian analysis PENDING

Priors stated and justified, prior-predictive baselines per rule, parameter recovery, sensitivity ladder: all above the expected standard, and the specification is now internally consistent on δ_O. The filter is disclosed. What decides this requirement is the model comparison the brief asks for by name — five LOO comparisons promised in §7, and in v2 only one was reported, by rank alone, in violation of the report's own reading rule. That is the gap to close when §8 is written.

8 — Mitigation MET

$M1$ is pre-registered, its geometric basis is argued independently of the fit — shortening S raises f_s/S and thins the stride branch while f_s/P is invariant, so overlap cannot reach the patch branch — and instantiated on the one-minute telemetry. That argument was never affected by the sign error and is what carries the requirement. The verdict is now *inconclusive* with both legs stated. *Two cautions for the rewrite:* the numbers quoted in §9 (0.44, 0.38, 0.964, 0.977) come from the previous fit and must be re-read from the new one; and the interpretation paragraph about $N = 1 + \lfloor (L - P)/S \rfloor$ still reads as an explanation of a *positive* slope — with the orientation fixed, a positive slope is now the predicted direction, so that paragraph should be reframed as a confound on the *estimate* rather than an excuse for a wrong sign.

9 — Agentic AI MET

Represent, monitor, explain and act are all present and grounded in the ontology, and the point that monitoring must sit upstream of tokenisation is the load-bearing one. Thinnest clause remains human-facing explanation: one sentence, no worked artefact or model-card entry shown.

10 — NIST AI RMF PARTIAL

Threat models, attack surfaces, the three named contexts and an honest residual-exposure statement are all present, and T4 now reads correctly. The standing gap is structural: GOVERN / MAP / MEASURE / MANAGE appear only as one column of `tab:riskRegister`. No prose defines them, no category or subcategory is cited (MAP 1.1, MEASURE 2.7, and so on), and GOVERN is never discussed at all. This is the clause most likely to be marked down on the Computer Security half, and it is cheap to fix: *two paragraphs organising §10 under the four functions, with subcategory citations, and the existing content redistributed under them.*

Priority order from here

1. Write the executive summary; delete `eight.tex:21-24`. *Thirty minutes.*

2. Re-run the fits and write §8, with θ_P, an $H3$-location verdict, the five LOO comparisons with their standard errors, and a convergence sentence you can keep.
3. Add the θ_{lock} row to `tab:bayesDecisions` *before* the refit.
4. Update the δ_O numbers in §9 and reframe the confound paragraph.
5. Restructure §10 under the four RMF functions.
6. Cheap wins if the words are there: the ε-neighbourhood sentences in §3, a `Forecast` class in the ontology, the phase count in §7, the twenty-two geometries listed, and either finishing or cutting Appendix E's planned extension.